

Python Programming

Time: 3 Hours

Max Marks: 70

Q.1 Attempt the following (Any FIVE)

(5 × 2 = 10 Marks)

1. Define Python interpreter.
 2. List any four built-in functions in Python.
 3. Explain the use of range() function.
 4. What is the difference between local and global variables?
 5. Define recursion with example.
 6. What is a constructor in Python?
 7. Explain the use of import statement.
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Q.2 Attempt the following (Any THREE)

(3 × 4 = 12 Marks)

1. Write a Python program to reverse a given number.
2. Explain different types of loops in Python with examples.
3. Write a Python program to count vowels in a given string.
4. Explain List methods:
 - append()
 - insert()
 - remove()
 - pop()
5. Write a Python program to find Fibonacci series up to n terms.

Q.3 Attempt the following (Any THREE)**(3 × 4 = 12 Marks)**

1. Explain Tuple and Dictionary with suitable examples.
 2. Write a Python program to check whether a string is palindrome or not.
 3. Explain function arguments in Python:
 - Default argument
 - Keyword argument
 - Variable length argument
 4. Write a program to read and write data into a file.
 5. Explain inheritance with example.
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Q.4 Attempt the following (Any THREE)**(3 × 6 = 18 Marks)**

1. Write a Python program to create a class Employee having:
 - Employee ID
 - Name
 - SalaryDisplay employee information.
 2. Explain exception handling mechanism in Python with example.
 3. Write a Python program using NumPy to perform array operations.
 4. Explain Tkinter widgets:
 - Label
 - Button
 - Entry
 - Text
 5. Write a Python program to store student details in SQLite database.
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Q.5 Attempt the following (Any TWO)

(2 × 8 = 16 Marks)

1. Develop a Python application for **Library Management System** using:
 - Classes
 - Functions
 - File Handling
2. Write a Python program to analyze student marks using Pandas:
 - Create DataFrame
 - Display records
 - Calculate average marks
3. Create a Tkinter GUI application for **Student Registration Form** with:
 - Name
 - Roll Number
 - Email
 - Submit Button
4. Explain Python database connectivity with SQLite and write a program to insert and display records.